Predicting What Makes Riders Happy in Ride-Sharing Services

A Machine Learning and Data Analysis Project

Sambit Roy

Python • Machine Learning • Data Analysis • Statistical Modeling

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GitHub Repository:

github.com/roysambit21/RideSharing-Rating-Analysis

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Predicting What makes Riders Happy in Ride Sharing Services:

1 Why This Project Was Chosen

This project was chosen because it offered a chance to work with a real dataset and apply data science ideas in a practical way. The main goal was to not only build a machine learning model but also to understand the entire workflow of a real data analysis project.

Through this project, I could show:

• Practical Python programming skills

• Data cleaning and preprocessing on actual survey data

• Exploratory Data Analysis (EDA) to find trends and patterns

• Feature engineering techniques to improve analysis

• Statistical thinking and interpretation of data

• Machine learning model building and evaluation

• Ability to extract meaningful business insights from data

• Professional documentation and presentation of the project

The dataset used in this project comes from a real ride-sharing survey collected from users, which makes the analysis more realistic and relevant to practical applications.

2 Project Background

Service_Rating has an essential part in the performance of ride-sharing applications. When users receive poor services on these apps, they tend to move to other similar ride-sharing services. For the majority of rides, Service_Rating usually does not depend on a single criterion but instead takes into account a number of other factors.

Some of those criteria can be:

- Wait time until the start of the ride
- Total cost of the ride
- Cleanliness of the car
- The behavior of the driver
- Convenience of ordering
- Passengers' security and comfort while driving

Thus, considering all these factors, the project aims at gaining insight into customers' experience based on data analysis and machine learning.

The main objectives of the project are:

- Analyzing the factors affecting customer satisfaction
- Forecasting Service_Rating based on machine learning models
- Developing useful insights that will be helpful for ride-sharing companies.

3 Objectives of the Project

Primary Goals

- Clean and pre-process the data from the survey data set
- Conduct EDA to find patterns and trends
- Learn about the factors that have the greatest impact on customer satisfaction
- Model different prediction models using Machine Learning algorithms
- Comparison of the performance of the different models

- Conduct feature importance analysis

Secondary Goals

- Presentation of knowledge about the whole process in machine learning
- Creation of an adequate project for applying to an MSc course in Data Science/Artificial Intelligence
- Application of practical skills in Python, data analysis, and machine learning

4 Data Collection Process

The information for this study was gathered through GOOGLE FORMS. This survey mainly revolved around the user experience regarding ride-sharing, which included the wait time, hygiene, booking process, cost, and satisfaction levels.

Survey Question / Variable	Description	Response Type / Scale Interpretation
Age	Age of respondent	Numerical / categorical age values
Gender	Gender identity of respondent	Male / Female
Ride Frequency	Frequency of ride-sharing usage	Daily, Weekly, Occasionally, Monthly, Rarely
Waiting Time	Average waiting time after ride request	Ordered category:1 = Less than 5 min 2 = 5–10 min 3 = 11–20 min 4 = 21–30 min 5 = More than 30 min
Distance Travelled	Typical ride distance	Ordered category:1 = 2–7 km 2 = 7–15 km 3 = 15–21 km 4 = 21–30 km 5 = More than 30 km
Driver Rating	Average rating of drivers encountered	Scale from 1–5 where:1 = Very Poor 5 = Excellent
Cleanliness of Vehicle	Frequency of clean and well-maintained vehicles	Scale from 1–5 where:1 = Never 5 = Always

Survey Question / Variable	Description	Response Type / Scale Interpretation
Ease of Booking	Ease of booking rides through app	Scale from 1–5 where:1 = Very easy 5 = Very difficult
Cost of Ride	Perceived affordability of ride-sharing services	Scale from 1–5 where:1 = Very affordable 5 = very expensive
Time of Day	Most frequent ride usage period	Morning / Afternoon / Evening / Night
Overall Satisfaction Rating	Overall satisfaction with ride-sharing services	Target variable:1 = Very Dissatisfied 2 = Dissatisfied 3 = Neutral 4 = Satisfied 5 = Very Satisfied

5 Data Description

Dataset Type

Survey-based ride-sharing experience dataset.

Approximate Size

~150 observations

Multiple categorical and ordinal variables

Examples of Features

Feature	Description
Age	Age group of respondent
Ride_Frequency	Frequency of ride-sharing usage
Waiting_Time	Time taken for pickup
Distance	Average distance travelled by User

Feature	Description
Driver_Rating	User rating of driver behavior
Cleanliness	Cleanliness rating
Ease_of_Booking	Booking convenience
Ride_Cost	Cost perception
Service_Rating	Final customer satisfaction

Target Variable

Overall Satisfaction Rating (Service_Rating)

6 Implementation

6.1 Importing Required Libraries

The project was implemented using Python libraries for:

- data manipulation,
- visualization,
- statistical analysis,
- and machine learning.

Major libraries used:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- SciPy

6.2 Loading and Understanding the Dataset

Dataset was loaded in Python using the Pandas package.

An initial investigation
n of the dataset was done in order to:

- find out how data is structured,
- check features,
- check for missing values,
- and find the dimensions of the dataset.

There are about:

- 150 observations of surveys
- various categorical and ordinal features
- a single feature denoting customer satisfaction. The survey includes features such as:
  - age,
  - gender,
  - ride frequency,
  - waiting time,
  - cleanliness,
  - ease of booking,
  - cost perception,
  - and satisfaction rating.

6.3 Data Cleaning

Data cleaning was done manually prior to analysis.

Columns that are unnecessary for the analysis, such as respondent names and time stamps, were deleted directly from the CSV file.

Once the dataset is loaded into Python, the following assessments are made:

- Search for missing values
- Detecting data types

Check for duplicates:
The cleaned dataset had:

- No missing values
- No duplicates
- All columns uniformly formatted

Thus, there was no need for any further data cleaning steps.

Checking Duplicates

The dataset was checked for duplicate observations using:

df.duplicated().sum()

No duplicate rows were identified in the dataset.

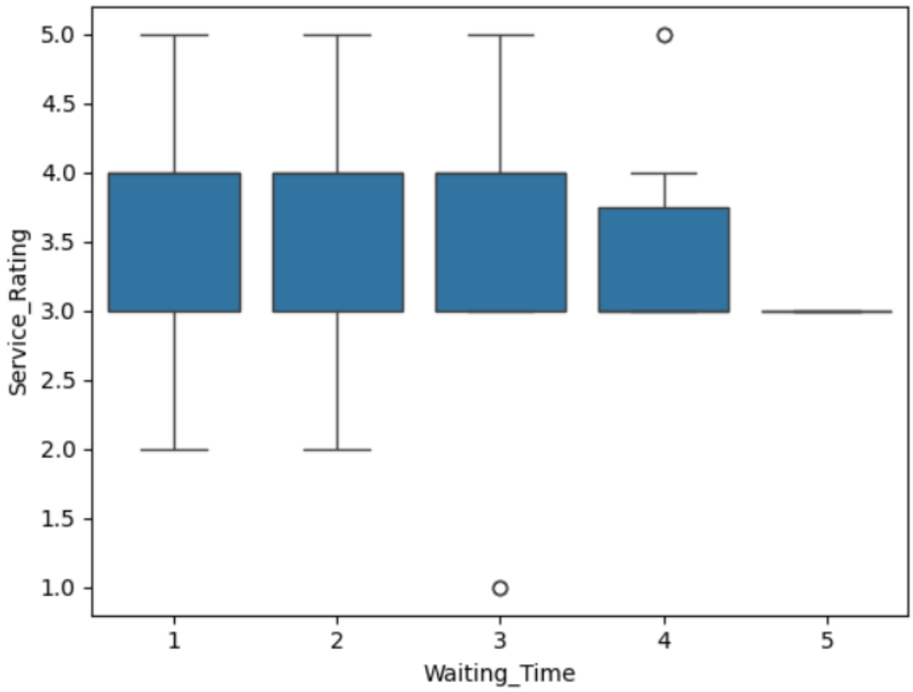
There were no possibilities of duplicate entries as the Google Form provided only one chance to respond from each individual participant.

The data set had no duplicate responses even after deleting personal columns such as Name and Email Address.

6.4 Exploratory Data Analysis

Q. 1 Does waiting time affect customer satisfaction?

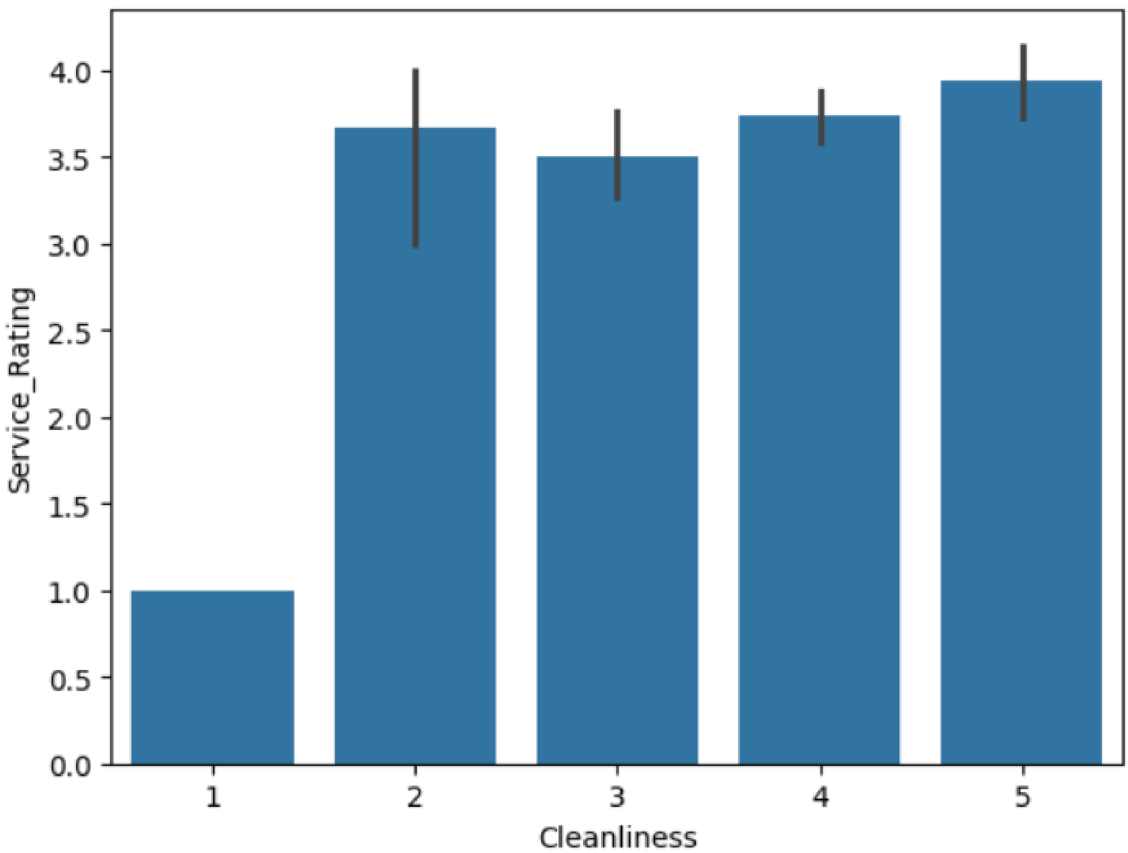
Visualization



Interpretation:

The boxplot shows that there is still considerable clustering of the Service_Rating scores in the high to moderate range regardless of the waiting time. While it may be that the variance in satisfaction is somewhat lower for longer waiting times, this pattern is not strongly visible in the data.

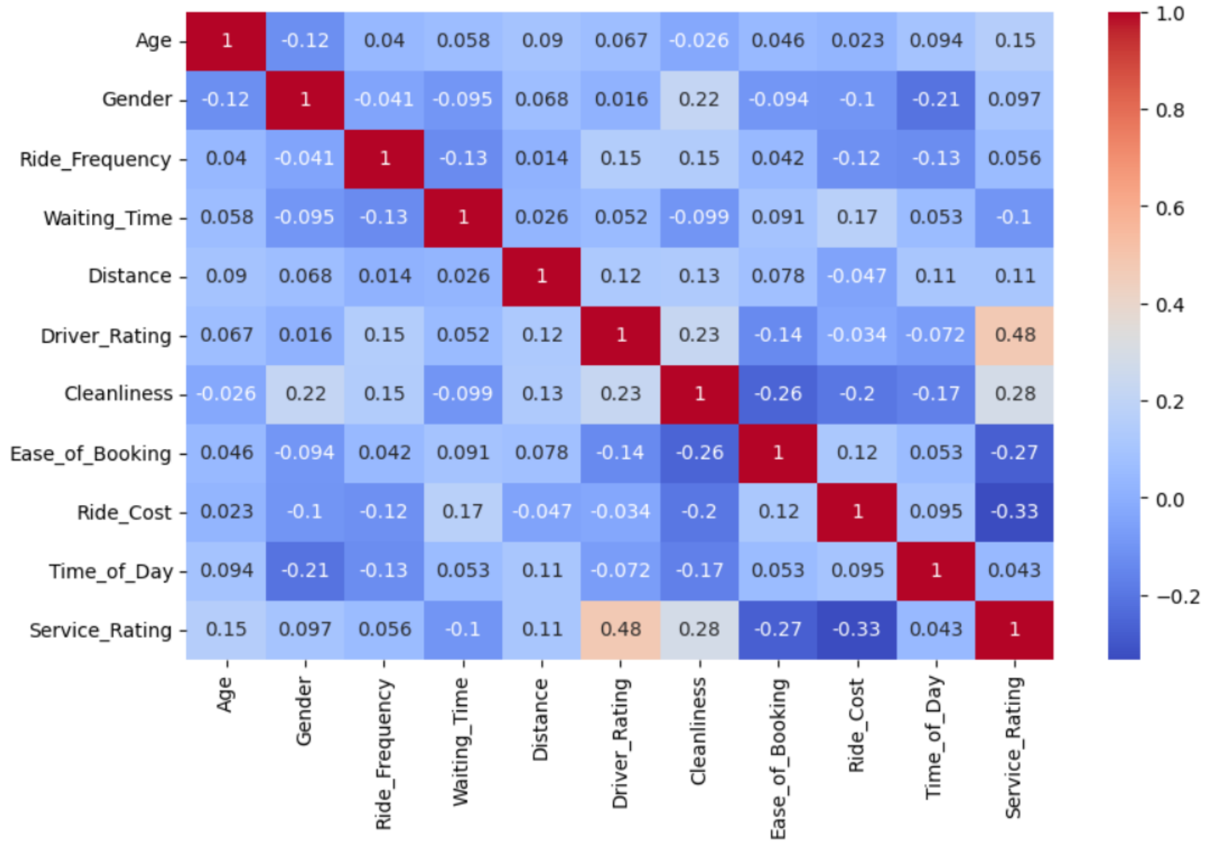
Q.2 Does cleanliness influence customer satisfaction?



Interpretation:

According to the visualization, there is a direct association between cleanliness of the vehicle and the level of customer satisfaction. The respondents who had cleaner vehicles tended to have higher scores on their satisfaction levels. Cleanliness levels that are at 4 or 5 show better results regarding customer satisfaction compared to lower levels.

Q.3 Which factors show strongest correlation?



Interpretation:

The correlation heatmap shows how different service features stack up against overall customer satisfaction. Out of everything, Driver_Rating stands out the most—it has the strongest positive link with Service_Rating (around 0.48). So, when people rate their drivers highly, they tend to rate the service itself higher too.

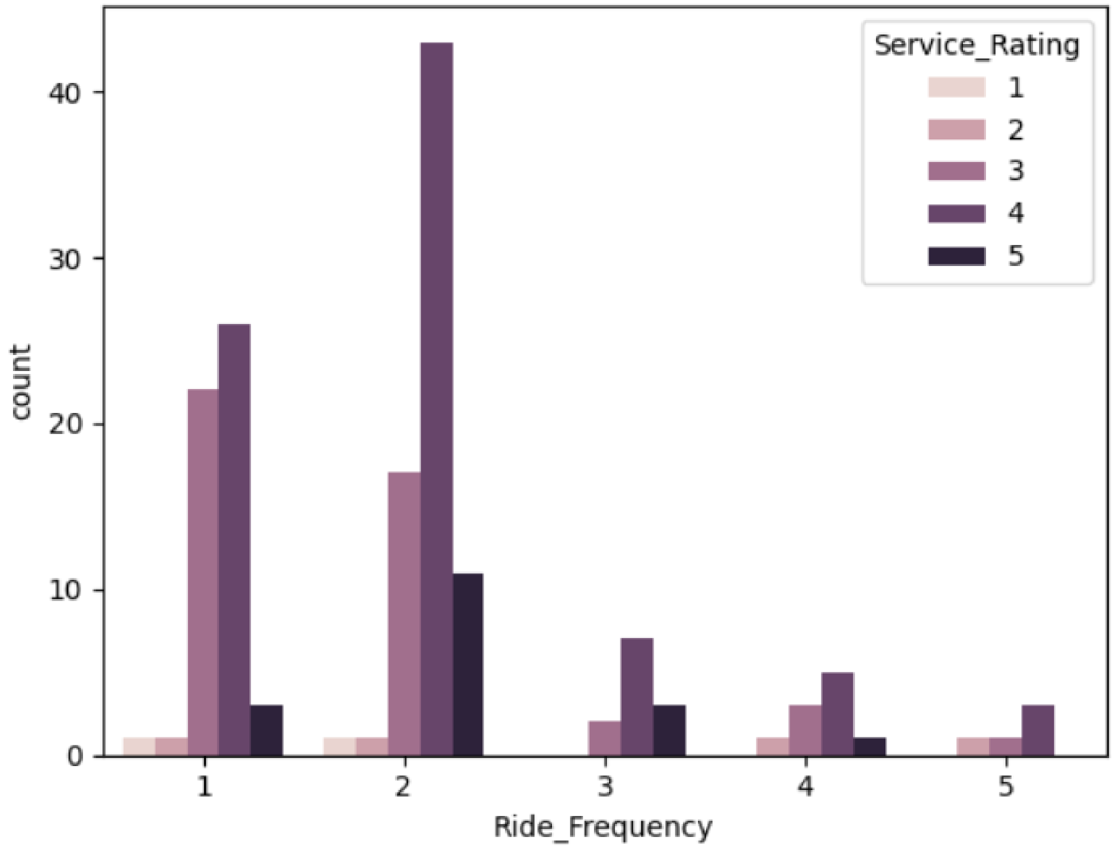
Cleanliness comes in next. It's not as big of a factor as the driver score, but there's still a clear connection: people are more satisfied when vehicles are clean and well-maintained.

On the other hand, Ride_Cost pulls things down. There's a moderate negative correlation here. Put simply, when people think the ride costs too much, their satisfaction drops.

Waiting_Time? That one doesn't seem to move the needle much. There's only a weak negative tie to satisfaction, so how long people wait doesn't play as big a role in how they feel about the service in this data.

Bottom line: things like driver behavior and cleanliness matter most for customer satisfaction. Stuff like cost matters too, just in the other direction, and wait times don't seem to change much for most riders.

Q.4 Does ride-sharing usage frequency influence customer satisfaction?



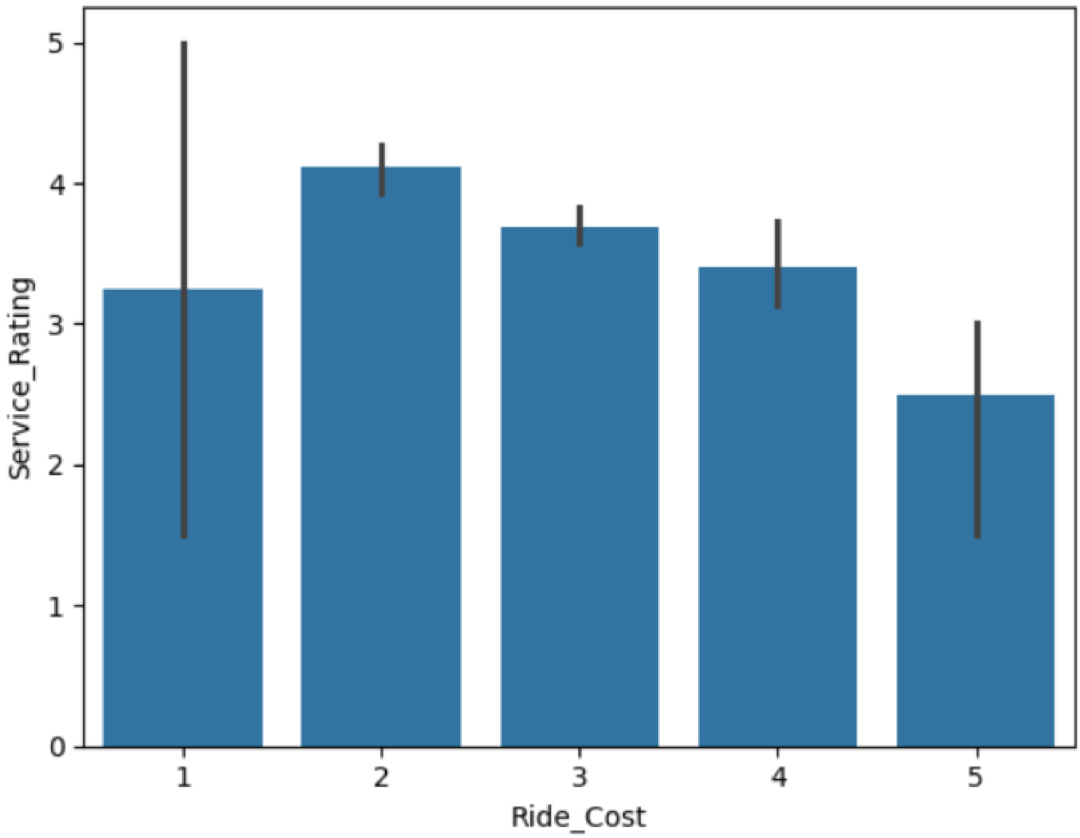
Interpretation:

This chart shows the distribution of Service_Rating based on the varying levels of ride-sharing usage frequency. Majority of the respondents fall under low usage-frequency groups, specifically "Rarely" and "Monthly."

In most usage-frequency groups, ratings are mostly centered on moderate to high ratings (3 and 4). This implies that the level of customer satisfaction does not change based on the ride-sharing usage frequency.

There is no clear trend showing that there is an impact on Service_Rating if the ride frequency is increased. This suggests that service rating can be determined more by service quality than by ride-sharing usage frequency.

Q.5 Does perceived ride cost influence overall customer satisfaction?



From the barplot, one can see the correlation between the level of customer satisfaction and the price level perception during the use of ride-sharing service. Customers who perceive moderate ride costs have higher average levels of service rating than those who perceive high ride costs.

This is evidenced by the fact that people with very high perception of ride costs have the least average levels of satisfaction among all other categories.

Nonetheless, from the high variations in some of the categories, it can be seen that there might be other factors that might contribute to customer satisfaction.

6.5 Statistical Analysis:

Hypothesis Testing:
Does driver rating significantly affect service rating?

Null Hypothesis (H0)

Driver rating and service rating are independent.

Alternative Hypothesis (H1)

Driver rating significantly affects service rating.

Result:

```
P-value: 8.389279504218833e-20
```

Interpretation: The output value of the chi-square test was found to be significantly smaller compared to the significance value of 0.05.

Thus, the Null Hypothesis (H_0) is rejected.

The outcome reveals that the Driver Rating and Service Rating variables are dependent on each other, thus indicating a statistically significant relationship between driver rating and customer satisfaction.

The conclusion correlates with the results obtained during the correlation study.

6.6 Machine Learning Model building:

Target Variable : Service_Rating

Train-Test Split:

The dataset was partitioned into the training set and the testing set using an 80:20 split ratio. The training set was used to learn the models, whereas the testing set was employed for testing the models' performance on unseen instances.

The random state is fixed to guarantee that the same partitions are obtained for each execution.

Model 1 — Logistic Regression

Objective

In this project we will build a baseline classification model, which we will use to predict Service_Rating ratings based upon ride-sharing service features.

Methodology

We used Logistic Regression as a linear classification technique to understand the predictors of Service_Rating levels from the features of the ride-sharing service.

We used a training dataset to train our model and an independent testing dataset to evaluate our model's predictive accuracy.

Why This Model Was Chosen

Logistic Regression is:

- Simple
- Easy to understand
- Computationally efficient
- Typically the base classification model chosen for building similar types of classification models

It can provide evidence that Service_Rating levels can be separated linearly from service-related features.

Results:

Model Performance

The Logistic Regression classification model performed at a rate of about +/- 61% accuracy on the test dataset. Logistic Regression performed comparatively well within the dominant satisfaction categories (i.e., there were more instances of ratings 3s and 4s in the dataset than any other combination of rating categories) at predicting these dominant categories. Conversely, logistic regression performed poorly with respect to those categories in the test dataset that had very few instances (i.e., these were the minority categories) that were predicted well (than that had many instances), and as a result the precision and recall rates for those categories were low.

Conclusion

The Logistic Regression classification model showed that it has moderate predictive capability when it comes to predicting Service_Rating using the characteristics of each ride-share service. Although the model predicted fairly well for service ratings that appeared frequently (i.e., for the majority category), it was not able to predict service ratings well (i.e., for the minority category) because of the class imbalance between the majority and minority classes and the limited number of instances within the dataset.

Despite these issues, Logistic Regression model was a reasonable/classification baseline model that could be used to compare similar types of service ratings across the dataset.

Model 2 — Decision Tree

Objective

To capture non-linear relationships between ride-sharing features and customer satisfaction.

Methodology

To express nonlinear relationships between ride-sharing characteristics and customer satisfaction.

Approach

A Decision Tree Classifier was implemented to model decision-based relationships among predictor variables. The classifier uses a recursive partitioning of the data to differentiate between features of importance to the splitting criteria and generate smaller data set sizes.

Why This Model Was Chosen

Decision trees:

- Are easy to read
- Can model anything that is not a linear relationship
- Generate a clear path from input to output

Because Decision Trees are particularly useful in understanding which features create the most significant decision boundaries.

Results

Model Performance

The testing dataset's Decision Tree model provided an overall accuracy of around 55 percent.

The Decision Tree model provided slightly better predictions when used to predict the dominant satisfaction category (i.e., rating of 4) while the prediction performance of the predominantly used Service_Rating categories was poorer than these types of ratings due to insufficient examples or instances in these Service_Rating categories.

Interpretation

When compared to Logistic Regression, the Decision Tree model produced less overall predictive accuracy on the test data that was not used for training.

The lesser predictive performance observed with the Decision Tree model may have resulted from it picking up very specific patterns in the training dataset on which it was trained, which may not generalize as well to the sample of unseen test data collected from a very small number of samples or instances in the test dataset, which is a common problem when using Decision Trees.

Overall, the model could still find moderate quantities of moderate Service_Rating patterns that occur often and will provide a good basis for predicting the Service_Rating.

Model 3 — Random Forest

Objective

Increase prediction accuracy with the help of an ensemble learning technique.

Methodology

Implement Random Forest as an ensemble classifier by utilizing several decision trees to construct the final classification. Predictions are generated through aggregated voting across multiple decision trees, helping reduce overfitting and improve generalization.

Random Forest Results

Model Performance

The Random Forest model yielded the highest accuracy of any of the implemented models with 71% accuracy on the testing data.

The model showed a strong ability to predict the dominant Service_Rating categories, especially the service rating 4 category which had a very high recall performance.

Interpretation

The Random Forest model performed better than both the Logistic Regression model and Decision Tree model in terms of its generalization ability on unseen test data. The Random Forest model is likely to have improved predictive accuracy through its ensemble based structure by combining the predictions from multiple decision trees and reducing the common overfitting tendencies found with using a single decision tree for prediction.

The Random Forest model found meaningful relationships between the Ride Share Service attributes and the Service_Rating levels, providing for better predictability.

However, the Random Forest model continues to struggle to predict the minor classes in the dataset due to the imbalanced nature of the data and limited number of observations of some of the Service_Rating satisfaction levels.

Model Comparison:

Model	Accuracy
Logistic Regression	61%
Decision Tree	55%
Random Forest	71%

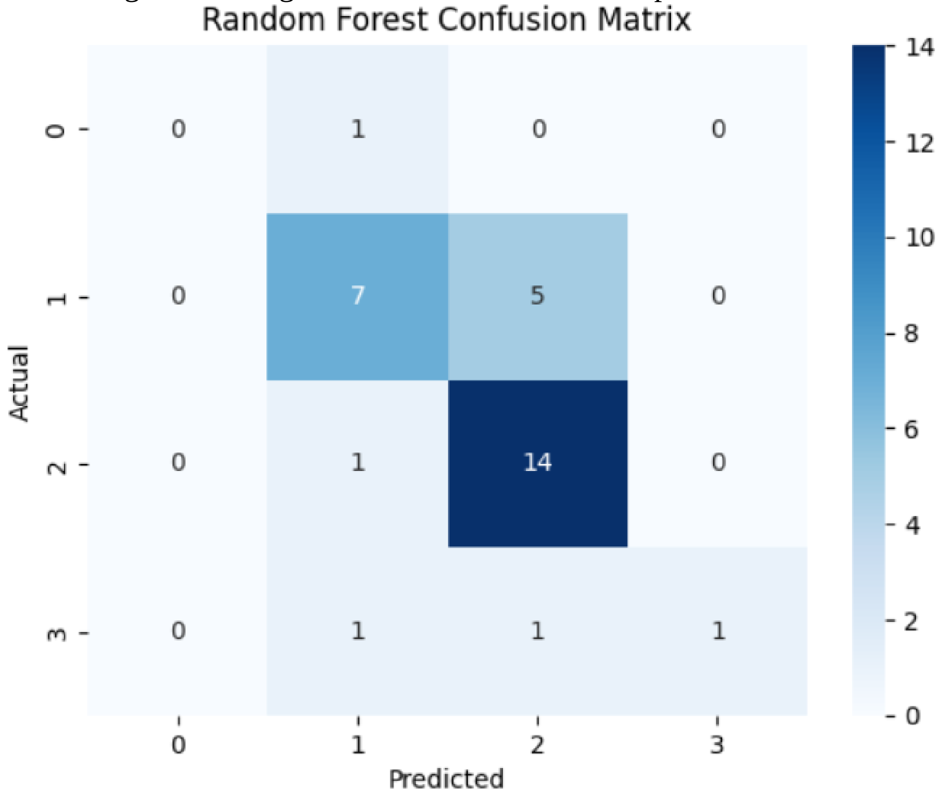
Comparative Analysis of Models

Among the machine learning models that were applied, Random Forest produced the best performance for accuracy when predicting based on the testing dataset.

Logistic Regression had a stable and moderate performance, therefore it provided a reasonable baseline model for predicting Service_Rating. Decision Tree had a lower degree of generalization and likely fit this model poorly because of over-fitting problems due to a small dataset.

Random Forest used the ensemble learning (combining models) approach to outperform the previous models by being able to more effectively handle how features interacted with each other and reduce the variance of an accurate prediction.

In summary, Random Forest was determined to be the most suitable model for predicting Service_Rating when using this dataset as the basis for prediction.



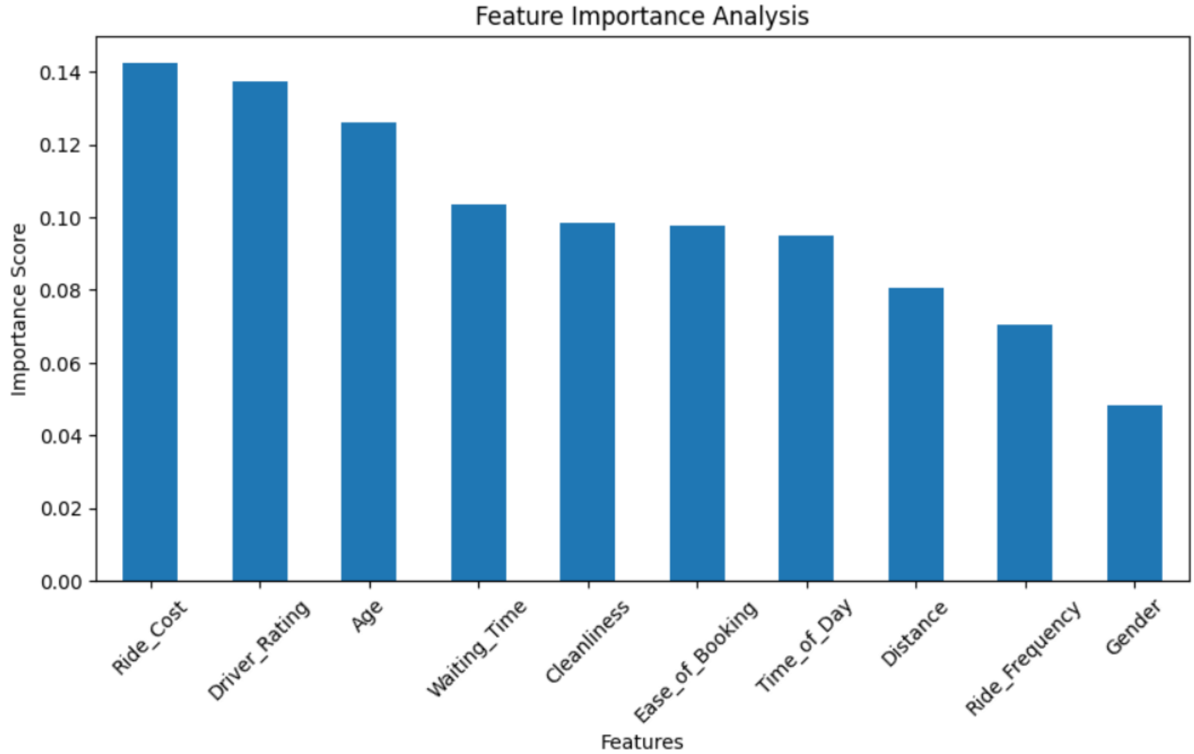
6.7 Feature Importance Analysis:

Objective

To determine which of the attributes of ride share services are the most important in predicting Service_Rating via Random Forest models.

Methodology

Utilizing the trained Random Forest algorithm, we can calculate feature importance for each independent variable in order to determine how each contributes to the prediction of Service_Rating. With feature importance calculations, we can analyse those attributes that influence the model decision making in greater detail than if we simply evaluated the output from the model alone.



Interpretation

The feature importance analysis indicates that Ride_Cost and Driver_Rating were the most influential variables affecting Service_Rating prediction within the Random Forest model.

This suggests that customer perception of ride affordability and driver experience play major roles in shaping overall satisfaction levels in ride-sharing services.

Variables such as Waiting_Time, Cleanliness, and Ease_of_Booking also demonstrated moderate predictive importance, indicating that operational and usability-related service factors contribute meaningfully toward customer experienceIn contrast, both Gender and Ride_Frequency have lower importance scores, indicating they are likely to influence Service_Rating predictions for this dataset less than many of the other features.

Overall, the findings for feature importance were consistent with the prior correlation analysis and the hypothesis testing, adding further support to the quality and consistency of the overall analysis performed.

Key Insight

The factors that influence Service_Rating for ride-sharing services are more related to cost and service quality than to demographic factors (such as gender and frequency of rides).

6.8 Overall Business Insights

1. Driver experience has the strongest influence on user satisfaction.

Using correlation, hypothesis testing and feature importance analysis, Driver_Rating has consistently been found to have the greatest impact on Service_Rating. This indicates that drivers' actions, professional behaviour and the experience of a customer during the ride have a significant impact on the level of satisfaction a user receives from the ride-sharing service.

2. User perception of ride cost significantly impacts overall user satisfaction.

In Random Forest models, the Ride_Cost has emerged as being the greatest determining factor in producing overall user satisfaction. Customers who perceived that they were paying a higher price for a ride compared to other riders reported being significantly less satisfied with their overall experience, suggesting that pricing strategy and affordability may have a large impact on long-term customer retention and loyalty to a platform.

3. Quality of service is rated more highly than operational characteristics.

Variables related to service quality and user experience, including Cleanliness, Driver Rating, and booking convenience, demonstrated stronger relationships with customer satisfaction than operational variables such as Ride Frequency and Waiting Time.

4. Waiting time showed weaker influence than initially expected

Traditionally believed to be a large variable in determining the experience of customers, the data has shown that there is a small correlation (or connection) between waiting time (Waiting_Time) and the satisfaction rating (Service_Rating) for rides in this dataset.

Customers may be able to endure some wait time, depending on how good their ride was and how much they spent on it.

5. The Random Forest algorithm performed better than other machine learning algorithms used in this study.

In analyzing the performance of 3 different Machine Learning algorithms, Random Forest was shown to have accurately predicted ~71% of the service ratings in this dataset and outperformed Logistic Regression and Decision Trees in their generalization ability.

Therefore, it is highly likely that ensemble learning techniques will yield better prediction of customer satisfaction trends for structured ride sharing survey data.

6. The number of dissatisfied and highly satisfied customers affected the ability of Machine Learning Models to accurately predict this group.

The small number of dissatisfied and highly satisfied riders in this dataset negatively impacted the machine learning algorithm's ability to predict the satisfaction of these two groups.

The use of a balanced dataset will achieve increased predictive accuracy for customer satisfaction models.

6.9 Overall Conclusion

This project has provided proof of all parts of a real-world data science project. The project covered: preprocessing data through cleaning it and Exploring Data Analysis (EDA) via visualizations of data; completing statistical tests on data through Hypothesis Testing; building from data to machine learning models to compare model performance based on predicted values for new data and comparing the different models used; understanding which features showed the most importance for new customers with predicting high levels of ride-sharing customer satisfaction.

Based on our analysis, we found that customer satisfaction with ride-sharing services was strongly influenced by service quality and ride cost perception, including service level factors, such as driver experience, affordability factors, and overall ride quality factors.

Random forest performed better than all other models and provided us valuable value through the identification of features related to the customer satisfaction levels of ride-sharing services.

7 Limitations of the Project

1 . Limited Dataset Size:

Our study data contained roughly one hundred fifty completed surveys. Although it was enough to demonstrate the complete workflow of machine learning, due in part to its limited scope, the overall size of the data may have resulted in a lower level of generalizability and robustness of the analytical findings and predictive models.

2. Class Imbalance in Satisfaction Categories

There were a number of Service_Rating categories where only a few surveys were completed for each of the extremely dissatisfied and very satisfied within the satisfaction list. This resulted in an imbalance in the number of classes for each group, thus negatively affecting both the performance level of our machine learning models as well as their predictive accuracy for the minority class of cases.

3. Collecting Survey-Based Data

Data was collected through Google Forms via self-reported surveys with the following potential influencing factors affecting the analysis.

- response bias;
- Subjective perceptions by the respondent; and
- Variability of user interpretation of rating scales,

So, however, the actual customer behaviour can only sometimes be replicated by survey response results.

4. Feature Scope Restrictiveness

The dataset is limited to only selected ride share experience factors (e.g., Wait Time, Cleanliness, Cost of Ride, and Driver Rating). There are numerous other ride share experience factors that may influence the ride share experience but were not collected. Examples of those other factors include, but are not limited to:

- Traffic conditions
- Surge pricing
- App performance
- Income level of customer
- Current weather conditions
- Ride safety incidents

Including those additional factors may allow for improvements in predictive performance and analytical depth.

5. Simplification Through Ordinal Encoding

Many categorical variables have been transformed into numeric ordinal scales during the preprocessing stage of the machine learning development workflow. While this transformation allows for machine learning implementation, it has the potential to over-simplify customer perceptions and take away from potential nuance in the response categories.

6. Moderate Predictive Accuracy

Despite the Random Forest Model showing noticeably superior performance compared to previously implemented models (~71% accuracy) overall predictive accuracy is currently moderate rather than exceptionally high indicating that there we will likely be additional unobservable influences on customer satisfaction that are outside of our modeling dataset.

7. Risk of Overfitting with Small Sample Sizes

Modelling such as Decision Tree exhibited decreased generalization performance on new data, which may imply an overfitting tendency due to the size of the current modelling dataset and consequently the limited number of observations available for training.

8. Limited Generalizability Outside of Survey Sample Population

The survey responses in this study come from a restricted participant sample size that may not provide an accurate representation of all ride sharing users within various geographical regions, demographic subgroups or different types of ride sharing service.

Final Limitation Statement

The project's success illustrates how data preprocessing, exploratory data analysis, statistical testing and machine learning model development, as well as the assignment of feature importance, can be demonstrated in a real-world example for use in predicting customer satisfaction.

8 Future Improvements

1. Larger Data Set

Future expansion efforts should focus on the size of the data set to increase ride-sharing customers involved in the survey. Increased size of data set would create more statistical validity of answers from users to allow more accurate prediction of data trends. Additionally, A larger dataset would improve the robustness, reliability, and generalizability of both statistical analysis and machine learning models.

2. Address Class Imbalance

Data set contained a number of less than enough responses to certain attributions of (non-user) user satisfaction (two ad non-user). Efforts to correct the class imbalance will include; gather balanced responses, resample to create balance, use class balancing methods (e.g. SMOTE) or other class balancing techniques. This corrective effort will increase predictive capability of the satisfaction class (non-user).

3. Additional Independent Variables

Future studies should consider adding new independent variables which could affect ride-sharing patron user satisfaction such as;

- surge pricing
- traffic conditions
- weather
- cancelled ride frequency
- application performance (optimizations)
- incidences of safety
- income of user
- real-time ride length

Including richer features may improve both analytical depth and model accuracy.

4. Testing Advanced Machine Learning Techniques

This project used much simpler models than those that we would utilize to demonstrate core machine learning concepts — i.e. the machine learning models that are foundational and are most appropriate for demonstrating machine learning concepts. We also plan to look at using more sophisticated models in future work, such as:

- XGBoost,
- LightGBM,
- Support Vector Machines, or
- Neural Networks.

These would likely allow for better predictive performance of our models and allow for capturing more complex relationships in our data.

5. Tuning Hyperparameters

The models that we trained in the examples above primarily used the default parameters for their training. We could improve the performance of these models by systematically tuning the hyperparameters, e.g.:

- Grid Search,
- Random Search, and
- Cross Validation.

Each could further improve the generalizability and accuracy of our predictive models.

6. Developing the Applications of the Models for Real-Time Prediction

Future versions of our project will include incorporating our trained models in:

- Web Applications,
- Dashboards, and
- Ride-Sharing Management Systems to provide customers with real-time predictions about their level of satisfaction and operational insights based on the use of these types of technologies.

7. Sentiment Analysis Integration

Future research can be done to add text-based reviews and feedback to current numerical survey responses to gain deeper insight into customer experience using Natural Language Processing (NLP) techniques for Sentiment Analysis.

8. Geographic and Platform-Based Analysis

Future studies can compare the satisfaction levels of customers in various cities, regions, or ride share platforms to provide better understanding of how customers behave differently based on geographic area and ride share platform use.

9 Optional Deployment Possibility

The Random Forest model could potentially be implemented as an interactive web-based application, utilizing frameworks such as Streamlit. This potential application would allow a user to provide various ride share experience variables like:

• time to wait
• driver rating
• cleanliness
• cost of ride
• ease of booking

and have those variables generate a prediction on customer satisfaction levels in real-time.

This form of deployment could aid ride share companies in tracking the experience of customers while identifying potential service quality problems more effectively.

10 Tools and Development Environment

This project was developed using the following tools and technologies:

- Python Programming Language
- Google Colab Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- SciPy
- GitHub for project hosting and version control

The implementation and experimentation were performed in a cloud-based Google Colab environment.

11 References

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